

A rigorous weak-coupling strategy for the two remaining inputs in the continuum $P(\varphi)_2$ cutoff problem

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Abstract

The corrected continuum LPPL theorem in `lppl.tex` has exactly two open inputs: a Hamiltonian gap uniform in the spatial cutoff and a uniform local second moment of the interaction density. This note gives a single source-backed strategy for both inputs in a precisely specified weak-coupling regime. The required source estimates are the cutoff-uniform cluster and polynomial-moment estimates of Glimm–Jaffe–Spencer. We state those estimates in the form actually needed, prove the Euclidean-slab-to-ground-state transfer, derive an explicit uniform bound for the local second moment, prove density of the cylinder vectors, and then derive the physical Hamiltonian gap by the spectral theorem. Thus the remaining work is not a new spectral-gap argument: it is a finite source-scope audit and a careful normalization check. No stochastic-quantisation gap or change of Dirichlet form occurs.

1 Regime and exact target

Fix a real semibounded polynomial

$$\mathcal{P}(\xi) = \sum_{n=0}^p a_n \xi^n, \quad p \text{ even}, \quad a_p > 0,$$

a free mass $m_0 > 0$, and a coupling $\lambda \geq 0$. Set $\mathcal{G}_0 := C_c^\infty(\mathbb{R}; [0, 1])$. For $g \in \mathcal{G}_0$, put

$$K_g = H_0(m_0) + \lambda W(g), \quad W(f) = \int_{\mathbb{R}} f(x) : \mathcal{P}(\varphi_0(x)) : dx,$$

and let $E_g = \inf \text{spec } K_g$, $H_g = K_g - E_g$. Its normalized positive ground state is Ω_g and $P_g = |\Omega_g\rangle\langle\Omega_g|$.

The weak-coupling regime is the following quantified statement:

$$0 \leq \frac{\lambda}{m_0^2} < \varepsilon_{\mathcal{P}}, \tag{WC}$$

where $\varepsilon_{\mathcal{P}} > 0$ is the constant furnished by the norm-decreasing cluster expansion in [1, Thms. 1.1.7–1.1.8]. The statement “weak coupling” below always means (WC); it never means an unspecified neighbourhood.

Define the exact test-function class

$$\mathcal{F}_2 := \{f \in C_c^\infty(\mathbb{R}; \mathbb{R}) : \|f\|_\infty \leq 1 \text{ and } \text{supp } f \subset I \text{ for some interval } I \text{ with } |I| \leq 2\}.$$

The two estimates to prove are

$$\text{spec } H_g \subseteq \{0\} \cup [\gamma_*, \infty) \quad \text{for every } g, \tag{G}$$

$$\sup_{g \in \mathcal{G}_0, f \in \mathcal{F}_2} \|\lambda W(f) \Omega_g\|^2 < \infty. \tag{M_2}$$

2 The published cutoff-uniform input

Let $\mu_0^{(2)}$ be the massive free Euclidean field measure on $\mathcal{S}'(\mathbb{R}^2)$. For every measurable compactly supported $h : \mathbb{R}^2 \rightarrow [0, 1]$, define

$$d\nu_h = Z_h^{-1} \exp \left[-\lambda \int_{\mathbb{R}^2} h(z) : \mathcal{P}(\Phi(z)) : dz \right] d\mu_0^{(2)}.$$

Glimm–Jaffe–Spencer use unit-square localizations and a norm $\|\cdot\|_{\text{CE}}$ on sharp-time polynomial observables. For a localized monomial of local degrees $N(\Delta)$ and smearing kernel w , their norm has the form

$$\|Q\|_{\text{CE}} = K_1 \left(\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \right) \|w\|_2, \quad (2.1)$$

and a general polynomial is assigned the sum of the norms of its localized pieces.

The source input is exactly:

Hypothesis 2.1 (GJS cutoff-uniform estimates). *Under (WC), there are $C_{\text{CE}} < \infty$ and $m_{\text{CE}} > 0$, independent of h , such that:*

1. *for every sharp-time polynomial Q in the class of [1, (1.1.5)–(1.1.9)],*

$$|\nu_h(Q)| \leq C_{\text{CE}} \|Q\|_{\text{CE}}; \quad (\text{CE1})$$

2. *if Q_1, Q_2 have unit-square localizations separated by a strip of width d , then*

$$|\nu_h(Q_1 Q_2) - \nu_h(Q_1) \nu_h(Q_2)| \leq C_{\text{CE}} e^{-m_{\text{CE}} d} \|Q_1\|_{\text{CE}} \|Q_2\|_{\text{CE}}. \quad (\text{CE2})$$

Estimate (CE2) is Theorem 1.1.7 of [1], whose statement explicitly says that its constants are independent of h, Q_1, Q_2 . Estimate (CE1) is Theorem 1.1.8; in the proof it is the $B = 1, n = 0$ case of Theorem 1.1.11, whose error estimate is explicitly uniform in h . The allowed cutoff in [1, (1.8)] is an arbitrary compactly supported function $0 \leq h \leq 1$, not merely a rectangle. These three source clauses are the load-bearing scope checks.

3 From a Euclidean slab to the cutoff ground state

Let Ω_0 be the free Fock vacuum. For $T > 0$ define

$$h_{T,g}(t, x) = \mathbf{1}_{[-T, T]}(t) g(x), \quad \nu_{T,g} := \nu_{h_{T,g}}.$$

The characteristic function is admissible in the measurable-cutoff formulation of [1, (1.8)]. In fact, immediately after (1.8) the source itself takes $h_j(t, x) = g(x) \chi_{[-j, j]}(t)$; no limiting argument from smooth time cutoffs is required.

Lemma 3.1 (one-time slab transfer). *For every bounded time-zero multiplication operator A ,*

$$\nu_{T,g}(A) = \langle \psi_{T,g}, A \psi_{T,g} \rangle, \quad \psi_{T,g} := \frac{e^{-TH_g} \Omega_0}{\|e^{-TH_g} \Omega_0\|},$$

and $\psi_{T,g} \rightarrow \Omega_g$ in Hilbert norm as $T \rightarrow \infty$.

Proof. The Feynman–Kac–Nelson formula gives

$$\nu_{T,g}(A) = \frac{\langle \Omega_0, e^{-TK_g} A e^{-TK_g} \Omega_0 \rangle}{\langle \Omega_0, e^{-2TK_g} \Omega_0 \rangle}.$$

The factors e^{-TE_g} cancel, giving the first identity. Positivity improvement in Q -space implies $c_g := \langle \Omega_g, \Omega_0 \rangle > 0$. The spectral theorem gives

$$e^{-TH_g} \Omega_0 = c_g \Omega_g + e^{-TH_g} (1 - P_g) \Omega_0 \longrightarrow c_g \Omega_g.$$

Indeed, the squared norm of the second term is

$$\int_{(0,\infty)} e^{-2TE} d\mu_g^\perp(E), \quad \mu_g^\perp(B) := \langle (1 - P_g) \Omega_0, \mathbf{1}_B(H_g) (1 - P_g) \Omega_0 \rangle,$$

which tends to zero by dominated convergence. Thus even isolation of the ground-state eigenvalue has not been used. The norm tends to c_g , proving the assertion without any gap assumption. $\square$

Lemma 3.2 (two-time slab transfer). *Let A, B be bounded time-zero multiplication operators and $t \geq 0$. If the insertions are at Euclidean times $-t/2$ and $t/2$, then*

$$\lim_{T \rightarrow \infty} \nu_{T,g}(A(-t/2)B(t/2)) = \langle \Omega_g, A e^{-tH_g} B \Omega_g \rangle.$$

The assertion remains true for polynomial A, B if

$$\sup_{T \geq t/2+1} \nu_{T,g}(|A(\pm t/2)|^4 + |B(\pm t/2)|^4) < \infty. \quad (3.1)$$

Proof. For bounded insertions, FKN and cancellation of the scalar energy give the exact quotient

$$\frac{\langle u_{T,t}, A e^{-tH_g} B u_{T,t} \rangle}{\|e^{-TH_g} \Omega_0\|^2}, \quad u_{T,t} = e^{-(T-t/2)H_g} \Omega_0.$$

Both $u_{T,t}$ and $e^{-TH_g} \Omega_0$ tend to $c_g \Omega_g$, so the quotient has the asserted limit.

For an unbounded polynomial A , put $A_R = A \mathbf{1}_{\{|A| \leq R\}}$, and similarly for B . The bounded result applies to A_R, B_R . By Hölder and Markov,

$$\begin{aligned} \nu_{T,g}(|AB| \mathbf{1}_{\{|A| > R\}}) &\leq \nu_{T,g}(|A|^4)^{1/4} \nu_{T,g}(|B|^4)^{1/4} \nu_{T,g}(|A| > R)^{1/2} \\ &\leq R^{-2} \nu_{T,g}(|A|^4)^{3/4} \nu_{T,g}(|B|^4)^{1/4}. \end{aligned}$$

The analogous bound holds with A, B exchanged. Hypothesis (3.1) makes both errors uniformly $O(R^{-2})$. Moreover $A_R \Omega_g \rightarrow A \Omega_g$ and $B_R \Omega_g \rightarrow B \Omega_g$ in norm, since the corresponding second moments are finite. The contraction property of e^{-tH_g} therefore gives convergence of the truncated matrix elements on the right. First take $T \rightarrow \infty$, then $R \rightarrow \infty$. $\square$

By (CE1), condition (3.1) holds for every fixed GJS polynomial: its fourth power is again a finite sum of observables of the form (1.1.5), and its CE-norm is finite.

4 Explicit derivation of the local second moment

Proposition 4.1 (CE1 implies (M_2)). *Hypothesis 2.1(CE1) implies (M_2) . One admissible bound is*

$$\|\lambda W(f) \Omega_g\|^2 \leq 32\lambda^2 C_{\text{CE}} K_1 K_2^{4p} (2p)! \left(\sum_{n=0}^p |a_n| \right)^2. \quad (4.1)$$

Proof. Write I for an interval of length at most two containing $\text{supp } f$. It meets at most four unit spatial cells. Expanding the square gives

$$W(f)^2 = \sum_{n,m=0}^p a_n a_m \iint f(x)f(y) : \varphi_0(x)^n :: \varphi_0(y)^m : dx dy. \quad (4.2)$$

Split each integral according to the at most $4^2 = 16$ ordered pairs of cells. In each localized piece the kernel is a restriction of $f \otimes f$, so

$$\|f \otimes f\|_2 = \|f\|_2^2 \leq |I| \|f\|_\infty^2 \leq 2.$$

The total local degree is $n + m \leq 2p$. Therefore

$$\prod_{\Delta} N(\Delta)! \leq (n + m)! \leq (2p)!, \quad \prod_{\Delta} K_2^{2N(\Delta)} = K_2^{2(n+m)} \leq K_2^{4p}.$$

Enlarge K_2 to $\max\{K_2, 1\}$ if necessary. Using (2.1), summing the sixteen pieces and then n, m proves

$$\|W(f)^2\|_{\text{CE}} \leq 32K_1 K_2^{4p} (2p)! \left(\sum_{n=0}^p |a_n| \right)^2. \quad (4.3)$$

Thus CE1 bounds $\nu_{T,g}(\lambda^2 W(f)^2)$ by the right side of (4.1), uniformly in T, g, f .

Here the equality

$$\|\lambda W(f) \psi_{T,g}\|^2 = \nu_{T,g}(\lambda^2 W(f)^2)$$

is also justified for the unbounded observable: apply the bounded FKN identity to

$$W(f)^2 \wedge R := \min\{W(f)^2, R\},$$

and let $R \uparrow \infty$ by monotone convergence. The CE1 bound makes the resulting quantity finite.

By Lemma 3.1, $\psi_{T,g} \rightarrow \Omega_g$. Since $W(f)$ is closed, the quadratic form $\xi \mapsto \|W(f)\xi\|^2$ is lower semicontinuous in Hilbert norm. Hence

$$\|\lambda W(f) \Omega_g\|^2 \leq \liminf_{T \rightarrow \infty} \|\lambda W(f) \psi_{T,g}\|^2,$$

and (4.1) follows. $\square$

The numerical constant 32 is deliberately crude; its purpose is to show that no dependence on the position of I , the profile g , or the derivatives of f is hidden in the argument.

5 Explicit derivation of the physical Hamiltonian gap

Let $\mathcal{C}$ be the complex unital algebra of finite polynomials in compactly smeared real sharp-time fields; its self-adjoint part $\mathcal{C}_{\text{sa}}$ consists of the real-valued polynomial cylinders.

Lemma 5.1 (density of polynomial cylinder vectors). *For each fixed g ,*

$$\overline{\{F \Omega_g : F \in \mathcal{C}\}} = \mathcal{H}_0.$$

Consequently $\{(1 - P_g)F \Omega_g : F \in \mathcal{C}\}$ has dense span in $(1 - P_g)\mathcal{H}_0$.

Proof. In Q -space let $d\mu_g = \Omega_g^2 d\mu_0$; strict positivity of Ω_g makes $U : L^2(\mu_g) \rightarrow L^2(\mu_0)$, $UF = F\Omega_g$, unitary.

The n -point bound following Theorem 1.1.8 of [1], followed by the slab transfer and truncation, gives for every compactly smeared real field $X = \varphi_0(f)$ a constant $b_f < \infty$ such that

$$\mathbb{E}_{\mu_g}[X^{2n}] \leq (2n)!b_f^{2n} \quad (n \geq 0). \quad (5.1)$$

Thus, for $0 < a < b_f^{-1}$,

$$\mathbb{E}_{\mu_g}[e^{a|X|}] \leq 2\mathbb{E}_{\mu_g}[\cosh(aX)] \leq 2 \sum_{n=0}^{\infty} (ab_f)^{2n} < \infty. \quad (5.2)$$

We now prove the finite-dimensional density assertion rather than invoke it. Let $Y = (\varphi_0(f_1), \dots, \varphi_0(f_d))$ and let ρ be its law under μ_g . By Hölder and (5.2), there is $\alpha > 0$ such that

$$\int_{\mathbb{R}^d} \exp\left(2\alpha \sum_{j=1}^d |y_j|\right) d\rho(y) < \infty. \quad (5.3)$$

Suppose $u \in L^2(\rho)$ is orthogonal to every polynomial. On the connected tube

$$\mathcal{T}_\alpha := \{z \in \mathbb{C}^d : |\operatorname{Re} z_j| < \alpha \text{ for } 1 \leq j \leq d\}$$

define

$$L(z) = \int_{\mathbb{R}^d} u(y) e^{z \cdot y} d\rho(y).$$

Cauchy–Schwarz and (5.3) give an integrable majorant on every compact subset of $\mathcal{T}_\alpha$; the same is true after multiplication by any monomial in y . Consequently differentiation under the integral is valid and L is holomorphic on $\mathcal{T}_\alpha$. Orthogonality gives $\partial^\beta L(0) = \int u(y) y^\beta d\rho(y) = 0$ for every multi-index β . Hence the Taylor series of L at zero vanishes. The identity theorem (applied successively in the d variables) gives $L = 0$ throughout $\mathcal{T}_\alpha$. In particular,

$$\int u(y) e^{i\xi \cdot y} d\rho(y) = 0 \quad (\xi \in \mathbb{R}^d).$$

Uniqueness of the Fourier transform of the finite complex measure $u d\rho$ implies $u = 0$ ρ -almost everywhere. Thus polynomials are dense in $L^2(\rho)$.

Choose a countable dense family $(f_j)_{j \geq 1}$ of real Schwartz functions. The coordinate maps $\varphi_0(f_j)$ generate the completed Gaussian cylinder sigma-algebra; because $\Omega_g > 0$ μ_0 -almost everywhere, μ_g and μ_0 are equivalent and have the same completed null sets. If $\mathcal{F}_N = \sigma(\varphi_0(f_1), \dots, \varphi_0(f_N))$, the martingale convergence theorem gives $\mathbb{E}_{\mu_g}[Z | \mathcal{F}_N] \rightarrow Z$ in $L^2(\mu_g)$ for every $Z \in L^2(\mu_g)$. Each conditional expectation is an L^2 function of the first N coordinates and hence, by the preceding paragraph, is an L^2 limit of polynomials in those coordinates. This proves density in $L^2(\mu_g)$. Applying U proves the first assertion, and applying $1 - P_g$ proves the second. $\square$

Proposition 5.2 (CE2 implies the uniform physical gap). *Hypothesis 2.1(CE2) implies*

$$\operatorname{spec} H_g \subseteq \{0\} \cup [m_{\text{CE}}, \infty) \quad \text{for every } g.$$

Proof. Fix $F \in \mathcal{C}_{\text{sa}}$ and decompose it into finitely many localized monomials as in [1, (1.1.9)]. Translate one copy by Euclidean time t . For every pair of localized summands, the two

collections of unit squares are separated by a horizontal strip of width at least $t - 2$. Summing CE2 over the pairs therefore gives, for $t \geq 2$ and $T \geq t/2 + 1$,

$$|\nu_{T,g}(F(-t/2)F(t/2)) - \nu_{T,g}(F(-t/2))\nu_{T,g}(F(t/2))| \leq C_{\text{CE}} e^{2m_{\text{CE}}} \|F\|_{\text{CE}}^2 e^{-m_{\text{CE}} t}. \quad (5.4)$$

The fourth-moment version of CE1 verifies the uniform-integrability hypothesis of Lemma 3.2. Letting $T \rightarrow \infty$ in (5.4) yields

$$\begin{aligned} 0 &\leq \langle (1 - P_g)F\Omega_g, e^{-tH_g}(1 - P_g)F\Omega_g \rangle \\ &\leq C_F e^{-m_{\text{CE}} t} \quad (t \geq 0). \end{aligned} \quad (5.5)$$

For $0 \leq t < 2$, the left side is at most $\|(1 - P_g)F\Omega_g\|^2 \leq C_{\text{CE}}\|F^2\|_{\text{CE}}$ by CE1 and the slab transfer. Thus one may take

$$C_F = e^{2m_{\text{CE}}} C_{\text{CE}} \max\{\|F\|_{\text{CE}}^2, \|F^2\|_{\text{CE}}\},$$

which is independent of g . In particular, the decay exponent is independent of both g and F .

For completeness, fix $0 < \varepsilon < m_{\text{CE}}$ and let

$$R_\varepsilon = \mathbf{1}_{[0, m_{\text{CE}} - \varepsilon]}(H_g)(1 - P_g).$$

The spectral theorem and (5.5) give

$$e^{-(m_{\text{CE}} - \varepsilon)t} \|R_\varepsilon(1 - P_g)F\Omega_g\|^2 \leq C_F e^{-m_{\text{CE}} t}.$$

Multiplication by $e^{(m_{\text{CE}} - \varepsilon)t}$ and $t \rightarrow \infty$ show that $R_\varepsilon(1 - P_g)F\Omega_g = 0$. Lemma 5.1 and boundedness of R_ε give $R_\varepsilon = 0$: indeed, every $G \in \mathcal{C}$ is $G = (G + G^*)/2 + i(G - G^*)/(2i)$, a complex linear combination of two elements of $\mathcal{C}_{\text{sa}}$. As $\varepsilon \downarrow 0$, these projections increase strongly to $\mathbf{1}_{[0, m_{\text{CE}}]}(H_g)(1 - P_g)$; hence that projection is zero, which proves the claimed spectral inclusion. $\square$

6 Execution order and falsification gates

The work should be carried out in this order.

1. **Freeze the theorem.** State the first unconditional result only for the family $K_g = H_0(m_0) + \lambda W(g)$ satisfying (WC). Do not write “away from criticality”; that is a larger, unproved region.
2. **Complete the source-scope audit.** Check in the proof of [1, Thms. 1.1.8 and 1.1.11] that every occurrence of the cutoff is bounded only by $0 \leq h \leq 1$. Record the page and equation for each use. Failure criterion: any constant depending on the diameter, shape, derivatives, or time extent of h invalidates the transfer.
3. **Check conventions.** Match the free covariance, Wick mass, coefficient λ , and the energy shift between [1] and `lpp1.tex`. If the source Wick mass is $\tilde{m} \neq m_0$, use the identity

$$:\xi^n:_{\tilde{m}} = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{2^k k! (n - 2k)!} \left(\frac{1}{2\pi} \log \frac{\tilde{m}}{m_0} \right)^k : \xi^{n-2k} :_{m_0}. \quad (6.1)$$

This follows by comparing the generating functions $\exp(t\xi - C_m(0)t^2/2)$ and using the finite covariance difference $C_{m_0}(0) - C_{\tilde{m}}(0) = (2\pi)^{-1} \log(\tilde{m}/m_0)$. Recompute all lower-degree coefficients before applying (WC); merely renaming the Wick mass is not valid.

4. **Prove the two transfers.** Insert Lemmas 3.1–3.2 into the cutoff framework. The positivity overlap $\langle \Omega_g, \Omega_0 \rangle > 0$ and the fourth-moment tail bound must be written explicitly; weak convergence alone is insufficient for polynomial insertions.
5. **Discharge (M_2) first.** Verify the observable-norm calculation (4.3) directly from [1, (1.1.7)–(1.1.9)]. This is a finite calculation and is the earliest decisive test of the route.
6. **Discharge (G).** Transfer CE2, prove Lemma 5.1, and apply the spectral projection argument in Proposition 5.2. Do not cite the infinite-volume particle mass theorem as a substitute: what is needed is the finite-cutoff estimate uniform in g .
7. **Assemble.** Substitute $\gamma = m_{\text{CE}}$ and the constant from (4.1) into Theorem 5.3 of `lppl.tex`. The conclusion is full-net local-norm convergence for the weak-coupling family, not merely convergence of Schwinger distributions.

7 Routes not to pursue

1. An unweighted stochastic-quantisation LSI has the wrong Dirichlet form and the wrong generator. It is unnecessary in the weak-coupling route above.
2. The physical gap cannot control the ultraviolet second moment through a double commutator; the resulting coincident-point Wick square diverges.
3. The infinite-volume mass gap of [1, Thm. 1.1.2] does not by itself imply a gap for every inhomogeneous spatial cutoff. The load-bearing input is the cutoff-uniform finite-volume bound (CE2).
4. No analytic continuation in λ should be used to enlarge (WC) without a uniform lower bound on the gap along the continuation path; analyticity of finite-volume quantities does not prevent a full-net gap from closing.

8 Precise status

Theorem 8.1 (closure conditional only on the quoted GJS estimates). *Assume Hypothesis 2.1 with the stated cutoff scope. Then the two hypotheses (G) and (M_2) of `lppl.tex` hold with constants independent of $g \in C_c^\infty(\mathbb{R}; [0, 1])$. Consequently the full spatial-cutoff ground-state net converges in norm on every local algebra in the weak-coupling regime (WC), with all conclusions of Theorem 5.3 of `lppl.tex`.*

Proof. Proposition 4.1 proves (M_2), Proposition 5.2 proves (G), and Theorem 5.3 of `lppl.tex` performs the assembly. $\square$

The only remaining uncertainty in this strategy is documentary rather than conceptual: the published $O(1)$ in the moment estimate must be certified as uniform over the entire cutoff class used in the slab transfer. Theorem 1.1.11 says “uniformly in h ”, and Theorem 1.1.8 is its $B = 1, n = 0$ specialization, so the expected outcome is positive. If the line-by-line audit confirms that reading, the two formerly open inputs are simultaneously closed in the regime (WC). No claim is made outside that regime.

References

- [1] J. Glimm, A. Jaffe, and T. Spencer, *The Wightman axioms and particle structure in the $\mathcal{P}(\varphi)_2$ quantum field model*, Ann. of Math. **100** (1974), 585–632; doi:10.2307/1970959.